

# Masatoshi Katabuchi

Associate Professor

Xishuangbanna Tropical Botanical Garden, Chinese Academy of Sciences

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## Research Interest

Community ecology, Tropical forest ecology, Functional traits, Global change biology, Bayesian and likelihood methods

## Education

**Tohoku University, Japan**, Supervisor: Tohru Nakashizuka

2012 **Ph.D. (Life Sciences)**, Graduate School of Life Sciences “Community assembly and dynamics in tropical rainforests on the basis of functional traits”.

2009 **M.S. (Life Sciences)**, Graduate School of Life Sciences, “Reproductive ecology of a common dioecious fig and its pollinating wasp in a tropical secondary forest in Sarawak, Malaysia”

2007 **B.S. (Biology)**, Department of Biology

## Academic & Research Positions

June 2019-

**Associate Professor**, Xishuangbanna Tropical Botanical Garden, Chinese Academy of Sciences, China

August 2017-February 2019

**Postdoctoral Research Associate**, Michigan State University, USA

January 2017-August 2017

**Visiting Scholar**, University of Florida, USA

March 2016-January 2017

**Postdoctoral Research Associate**, University of Florida, USA

February 2014-February 2016

**JSPS Postdoctoral Fellow for Research Abroad**, University of Florida, USA

September 2012-February 2014

**Postdoctoral Research Fellow**, Xishuangbanna Tropical Botanical Garden, Chinese Academy of Sciences, China

April-August 2012

**Postdoctoral Research Fellow**, Tohoku University, Japan

April 2010-March 2012

**JSPS Research Fellow DC2**, Tohoku University, Japan

## Grants & Fellowships

2023 Xishuangbanna State Rainforest Talent Support Program 360,000 RMB

2022 ZiHui (Wisdom) Yunnan Program (202203AM140026) 120,000 RMB

2021 National Natural Science Foundation of China/RGC (3211101015) PIs: Louise Ashton, Akihiro Nakamura. Co-Is: Jin Wu, Min Cao, Masatoshi Katabuchi.

2020 Chinese Academy of Sciences President's International Fellowship Initiative (2020FYB0003) 300,000 RMB

2014 JSPS Postdoctoral Fellowship for Research Abroad (no. 25-504) 10,512,000 JPY

2013 Chinese Academy of Sciences Fellowships for Young International Scientists (no. 2013Y1SB0002) 165,000 RMB

2010 Grant-in-Aid for Scientific Research for JSPS Research Fellowship (no. 22-7035) 1,400,000 JPY

## Collaborator Named on Foreign Grants

2021 The Japan Society for the Promotion of Science Grant-in-Aid for Scientific Research (21H02248). PI: Yoshiko Iida. Co-PIs: Wakana Azuma, Mai Kamakura, Takashi Kohyama. External Scientific Collaborator: Masatoshi Katabuchi

## Publications

1. Zhou, C., Nakamura, A., Song, X. & **Katabuchi, M.** Artificial light at night (ALAN) influences understory plant traits through ecological processes: A two-year experiment in a rubber plantation in China. *EcoEvoRxiv* (2023) [doi].
2. Phouthavong, K., **Katabuchi, M.**, Nakamura, A., Cheng, X. & Cao, M. Interspecific and intraspecific adaptations of pteridophyte leaf traits in limestone and non-limestone forests of monsoon tropical regions of southwest China. *Journal of Plant Ecology* rtad026 (2023) [doi].
3. Maenpuen, P., **Katabuchi, M.**, Onoda, Y., Zhou, C., Zhang, J.-L. & Chen, Y.-J. Sources and consequences of mismatch between leaf disc and whole-leaf leaf mass per area (LMA). *American Journal of Botany* 109, 1242–1250 (2022) [doi].
4. Chen, Y.-J., Maenpuen, P., Zhang, Y.-J., Barai, K., **Katabuchi, M.**, Gao, H., Kaewkamol, S., Tao, L.-B. & Zhang, J.-L. Quantifying vulnerability to embolism in tropical trees and lianas using five methods: Can discrepancies be explained by xylem structural traits? *New Phytologist* 229, 805–819 (2021) [doi].
5. Chen, Y.-J., Choat, B., Sterck, F., Maenpuen, P., **Katabuchi, M.**, Zhang, S.-B., Tomlinson, K. W., Oliveira, R. S., Zhang, Y.-J., Shen, J.-X., Cao, K.-F. & Jansen, S. Hydraulic prediction of drought-induced plant dieback and top-kill depends on leaf habit and growth form. *Ecology Letters* 24, 2350–2363 (2021) [doi].
6. Swenson, N. G., Hulshof, C. M., **Katabuchi, M.** & Enquist, B. J. Long-term shifts in the functional composition and diversity of a tropical dry forest: A 30-yr study. *Ecological Monographs* e01408 (2020) [doi].
7. Sreekar, R., **Katabuchi, M.**, Nakamura, A., Corlett, R. T., Slik, J. W. F., Fletcher, C., He, F., Weiblen, G. D., Shen, G., Xu, H., Sun, I.-F., Cao, K., Ma, K., Chang, L.-W., Cao, M., Jiang, M., Gunatilleke, I. A. U. N., Ong, P., Yap, S., Gunatilleke, C. V. S., Novotny, V., Brockelman, W. Y., Xiang, W., Mi, X., Li, X., Wang, X., Qiao, X., Li, Y., Tan, S., Condit, R., Harrison, R. D. & Koh, L. P. Spatial scale changes the relationship between beta diversity, species richness and latitude. *Open Science* 5, 181168 (2018) [doi].
8. Johnson, D. J., Needham, J., Xu, C., Massoud, E. C., Davies, S. J., Anderson-Teixeira, K. J., Bunyavejchewin, S., Chambers, J. Q., Chang-Yang, C.-H., Chiang, J.-M., Chuyong, G. B., Condit, R., Cordell, S., Fletcher, C., Giardina, C. P., Giambelluca, T. W., Gunatilleke, N., Gunatilleke, S., Hsieh, C.-F., Hubbell, S., Inman-Narahari, F., Kassim, A. R., **Katabuchi, M.**, Kenfack, D., Litton, C. M., Lum, S., Mohamad, M., Nasardin, M., Ong, P. S., Ostertag, R., Sack, L., Swenson, N. G., Sun, I. F., Tan, S., Thomas, D. W., Thompson, J., Umaña, M. N., Uriarte, M., Valencia, R., Yap, S., Zimmerman, J., McDowell, N. G. & McMahon, S. M. Climate sensitive size-dependent survival in tropical trees. *Nature Ecology & Evolution* 1 (2018) [doi].
9. Osnas, J. L. D., **Katabuchi, M.**, Kitajima, K., Wright, S. J., Reich, P. B., Van Bael, S. A., Kraft, N. J. B., Samaniego, M. J., Pacala, S. W. & Lichstein, J. W. Divergent drivers of leaf trait variation within species, among species, and among functional groups. *Proceedings of the National Academy of Sciences of the United States of America* 115, 5480–5485 (2018) [doi].
10. Li, Y., Shipley, B., Price, J. N., Dantas, V. de L., Tamme, R., Westoby, M., Siefert, A., Schamp, B. S., Spasojevic, M. J., Jung, V., Laughlin, D. C., Richardson, S. J., Bagousse-Pinguet, Y. L., Schöb, C., Gazol, A., Prentice, H. C., Gross, N., Overton, J., Cianciaruso, M. V., Louault, F., Kamiyama, C., Nakashizuka, T., Hikosaka, K., Sasaki, T., **Katabuchi, M.**, Frenette Dussault, C., Gaucherand, S., Chen, N., Vandewalle, M. & Batalha, M. A. Habitat filtering determines the functional niche occupancy of plant communities worldwide. *Journal of Ecology* 3, 808–809 (2017) [doi].
11. **Katabuchi, M.**, Wright, S. J., Swenson, N. G., Feeley, K. J., Condit, R., Hubbell, S. P. & Davies, S. J. Contrasting outcomes of species- and community-level analyses of the temporal consistency of functional composition. *Ecology* 98, 2273–2280 (2017) [doi].
12. **Katabuchi, M.**, Kitajima, K., Wright, S. J., Van Bael, S. A., Osnas, J. L. D. & Lichstein, J. W. Decomposing leaf mass into photosynthetic and structural components explains divergent patterns of trait variation within and among plant species. *bioRxiv* (2017) [doi].

13. Siefert, A., Violle, C., Chalmandrier, L., Albert, C. H., Taudiere, A., Fajardo, A., Aarssen, L. W., Baraloto, C., Carlucci, M. B., Cianciaruso, M. V., de L. Dantas, V., de Bello, F., Duarte, L. D. S., Fonseca, C. R., Freschet, G. T., Gaucherand, S., Gross, N., Hikosaka, K., Jackson, B., Jung, V., Kamiyama, C., **Katabuchi, M.**, Kembel, S. W., Kichenin, E., Kraft, N. J. B., Lagerström, A., Bagousse-Pinguet, Y. L., Li, Y., Mason, N., Messier, J., Nakashizuka, T., Overton, J. M., Peltzer, D. A., Pérez-Ramos, I. M., Pillar, V. D., Prentice, H. C., Richardson, S., Sasaki, T., Schamp, B. S., Schöb, C., Shipley, B., Sundqvist, M., Sykes, M. T., Vandewalle, M. & Wardle, D. A. A global meta-analysis of the relative extent of intraspecific trait variation in plant communities. *Ecology Letters* 18, 1406–1419 (2015) [\[doi\]](#).
14. **Katabuchi, M.** LeafArea: An R package for rapid digital image analysis of leaf area. *Ecological Research* 30, 1073–1077 (2015) [\[doi\]](#).
15. Ashton, L. A., Nakamura, A., Basset, Y., Burwell, C. J., Cao, M., Eastwood, R., Odell, E., de Oliveira, E. G., Hurley, K., **Katabuchi, M.**, Maunsell, S., Mcbroom, J., Schmidl, J., Sun, Z., Tang, Y., Whitaker, T., Laidlaw, M. J., Mcdonald, W. J. F. & Kitching, R. L. Vertical stratification of moths across elevation and latitude. *Journal of Biogeography* 43, 59–69 (2016) [\[doi\]](#).
16. Nakamura, A., Burwell, C. J., Ashton, L. A., Laidlaw, M. J., **Katabuchi, M.** & Kitching, R. L. Identifying indicator species of elevation: Comparing the utility of woody plants, ants and moths for long-term monitoring. *Austral Ecology* 41, 179–188 (2016) [\[doi\]](#).
17. Nakamura, A., Burwell, C. J., Lambkin, C. L., **Katabuchi, M.**, Mcdougall, A., Raven, R. J. & Neldner, V. J. The role of human disturbance in island biogeography of arthropods and plants: An information theoretic approach. *Journal of Biogeography* 42, 1406–1417 (2015) [\[doi\]](#).
18. Deng, X., Mohandass, D., **Katabuchi, M.**, Hughes, A. C. & Roubik, D. W. Impact of Striped-Squirrel Nectar-Robbing Behaviour on Gender Fitness in *Alpinia Roxburghii* Sweet (Zingiberaceae). *PLoS ONE* 10, (2015) [\[doi\]](#).
19. Hikosaka, K., Sasaki, T., Kamiyama, C., **Katabuchi, M.**, Oikawa, S., Shimazaki, M., Kimura, K. & Nakashizuka, T. Understanding of species niche, coexistence and extinction based on functional traits : Approaches from community and physiological ecology for subalpine moorland plant communities [in Japanese]. *Chikyu-Kankyo* 19, 33–46 (2014).
20. Kamiyama, C., **Katabuchi, M.**, Sasaki, T., Shimazaki, M., Nakashizuka, T. & Hikosaka, K. Leaf-trait responses to environmental gradients in moorland communities: Contribution of intraspecific variation, species replacement and functional group replacement. *Ecol Res* 29, 607–617 (2014) [\[doi\]](#).
21. Sasaki, T., **Katabuchi, M.**, Kamiyama, C., Shimazaki, M., Nakashizuka, T. & Hikosaka, K. Vulnerability of moorland plant communities to environmental change: Consequences of realistic species loss on functional diversity. *Journal of Applied Ecology* 51, 299–308 (2014) [\[doi\]](#).
22. Cádiz, A., Nagata, N., **Katabuchi, M.**, DÍAZ, L. M., ECHENIQUE-DÍAZ, L. M., Akashi, H. D., Makino, T. & Kawata, M. Relative importance of habitat use, range expansion, and speciation in local species diversity of anolis lizards in cuba. *Ecosphere* 4, (2013) [\[doi\]](#).
23. Asanok, L., Marod, D., Duengkae, P., Pranamkol, U., Kurokawa, H., Aiba, M., **Katabuchi, M.** & Nakashizuka, T. Relationships between functional traits and the ability of forest tree species to reestablish in secondary forest and enrichment plantations in the uplands of northern Thailand. *Forest Ecology and Management* 296, 9–23 (2013) [\[doi\]](#).
24. Sasaki, T., **Katabuchi, M.**, Kamiyama, C., Shimazaki, M., Nakashizuka, T. & Hikosaka, K. Variations in species composition of moorland plant communities along environmental gradients within a subalpine zone in northern japan. *Wetlands* 33, 269–277 (2013) [\[doi\]](#).
25. Aiba, M., **Katabuchi, M.**, Takafumi, H., Matsuzaki, S. I. S., Sasaki, T. & Hiura, T. Robustness of trait distribution metrics for community assembly studies under the uncertainties of assembly processes. *Ecology* 94, 2873–2885 (2013) [\[doi\]](#).
26. Sasaki, T., **Katabuchi, M.**, Kamiyama, C., Shimazaki, M., Nakashizuka, T. & Hikosaka, K. Nestedness and niche-based species loss in moorland plant communities. *Oikos* 121, 1783–1790 (2012) [\[doi\]](#).
27. Sasaki, T., **Katabuchi, M.**, Kamiyama, C., Shimazaki, M., Nakashizuka, T. & Hikosaka, K. Diversity partitioning of moorland plant communities across hierarchical spatial scales. *Biodiversity and Conservation* 21, 1577–1588 (2012) [\[doi\]](#).
28. **Katabuchi, M.**, Kurokawa, H., Davies, S. J., Tan, S. & Nakashizuka, T. Soil resource availability shapes community trait structure in a species-rich dipterocarp forest. *Journal of Ecology* 100, 643–651 (2012) [\[doi\]](#).

29. **Katabuchi, M.**, Harrison, R. D. & Nakashizuka, T. Documenting the effect of foundress number in a dioecious fig, *Ficus Fistulosa*, in Malaysia. ***Biotropica*** 40, 457–461 (2008) [\[doi\]](#).
30. **Katabuchi, M.**, Isagi, Y. & Nakashizuka, T. Development of 17 microsatellite markers for *Ceratosolen Constrictus*, the pollinating fig wasp of *Ficus Fistulosa*. ***Molecular Ecology Resources*** 8, 840–842 (2008) [\[doi\]](#).

## Book

1. Hikosaka K., Sasaki T., Kamiyama C., **Katabuchi M.**, Oikawa S., Shimazaki M., Kimura H. and Nakashizuka T. (2016), Trait-Based Approaches for Understanding Species Niche, Coexistence, and Functional Diversity in Subalpine Moorlands. Structure and Function of Mountain Ecosystems in Japan.

## Software

- 2020 **ztpLn**: Zero-Truncated Poisson Lognormal Distribution, [R CRAN package](#)
- 2015 **LeafArea**: Rapid Digital Image Analysis of Leaf Area, [R CRAN package](#)
- 2015 **mglmN**: Model Averaging from Multivariate GLM with Null Models, R CRAN package [R CRAN package](#)

## Teaching Experience

- 2019- How do we measure and study biological diversity?, Advanced Field course in Ecology and Conservation, Xishuangbanna Tropical Botanical Garden, Chinese Academy of Sciences, China
- 2013 Experimental Design and Data Analysis, ATBC Asia-Pacific chapter, Banda Aceh, Indonesia
- 2013 R and generalized linear models, Xishuangbanna Tropical Botanical Garden, Chinese Academy of Sciences, China

## Service

- 2015-2017 Working Group Member of INNGE, International Network of Next-Generation Ecologists
- 2015-2017 INNGE manager, The Japan Society of Tropical Ecology
- 2013- Statistical editor for Lepcey, The Journal of Tropical Asian Entomology
- Reviewer for journals (~ 12 per year):

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|---------------------------------|-----------------------------------|
| - AoB Plants                    | - Journal of Ecology              |
| - Biotropica                    | - Journal of Plant Research       |
| - BMC Evolutionary Biology      | - Journal of Sustainable Forestry |
| - Ecological Research           | - Journal of Vegetation Science   |
| - Ecology Letters               | - Nature Communications           |
| - Ecosphere                     | - Oecologia                       |
| - Forest Ecology and Management | - Oikos                           |
| - Frontiers in Plant Science    | - Plant Ecology                   |
| - Functional Ecology            | - PLOS ONE                        |
| - Journal of Applied Ecology    | - Scientific Reports              |

## Awards and Prizes

- 2012 Dean's award for excellence, Graduate School of Life Sciences (Doctoral Program), Tohoku University
- 2009 Dean's award for excellence, Graduate School of Life Sciences (Master's Program), Tohoku University

## Invited Seminars

- 2021 Introduction to visualization using ggplot2, International Institute of Tropical Agriculture, Kenya
- 2018 A quantitative model for divergent drivers of leaf trait variation within and among plant species, Xishuangbanna Tropical Botanical Garden, China
- 2018 Species coexistence and functional traits. Graduate School of Life Sciences, Tohoku University, Japan
- 2018 Species coexistence and functional traits. Graduate School of Environment and Information Sciences, Yokoyama National University, Japan
- 2016 Decomposing leaf mass into photosynthetic and structural components explains divergent patterns of trait variation within and among plant species. Department of Ecology and Evolutionary Biology, Princeton University, NJ
- 2013 Additive partitioning of functional diversity in plant communities, Xishuangbanna Tropical Botanical Garden, China

## International Meetings

### Invited

- 2015 **Masatoshi Katabuchi**, “Future perspective for functional trait research”, Organized session on “Achievements and outstanding questions for trait-based forest ecology”, 62nd ESJ Annual meeting, Kagoshima, Japan
- 2014 **Masatoshi Katabuchi (Organizer)**, “Seedling dynamics and its consequence to community assembly processes in a tropical forest”, Symposium on “Integrating the structure and dynamics of tree communities to uncover the mechanisms underlying species coexistence”, 61st ESJ Annual meeting, Hiroshima, Japan
- 2011 **Masatoshi Katabuchi** and Hiroko Kurokawa “Community assembly and demography of tropical tree species”, Workshop on “Life history regulation of forest trees: towards cross-biome analysis”, Sapporo, Japan

### Submitted

- 2019 **Masatoshi Katabuchi**, Kaito Umemura, Thomas Koffel, Litchman Elena, Klausmeier A Christopher, “Combining metacommunity and trait-based approaches predicts species abundances distributions in biological communities”, 66th ESJ Annual meeting, Kobe, Japan
- 2015 **Masatoshi Katabuchi**, Kaoru Kitajima, Joseph Wright, Jeanne Osnas and Jeremy Lichstein “Decomposing leaf mass into photosynthetic and structural components explains diverge patterns of trait variation within vs. among plant communities”, 100th ESA Annual meeting, Baltimore, USA
- 2014 **Masatoshi Katabuchi**, Takehiro Sasaki, Chiho Kamiyama, Masaya Shimazaki, Tohru Nakashizuka and Kouki Hikosaka “Intraspecific trait variations affect spatial patterns of functional diversity among moorland plant communities”, 99th ESA Annual meeting, Sacramento, USA.
- 2013 **Masatoshi Katabuchi**, Sylvester Tan and Tohru Nakashizuka “Negative density dependence reduces survival rates of common tree species in a species-rich tropical rain forest”, ATBC, San Jose, Costa Rica
- 2012 **Masatoshi Katabuchi**, Hiroko Kurokawa, Stuart Davies, Sylvester Tan and Tohru Nakashizuka “Soil resource availability shapes community trait structure in a species-rich dipterocarp forest”, ATBC Asia-Pacific chapter, XTBG, China
- 2011 **Masatoshi Katabuchi**, Hiroko Kurokawa, Sylvester Tan, Stuart J. Davies, and Tohru Nakashizuka, “Non-random abundance distributions of a tropical tree community: evidence from traits correlation at a local scale”. ATBC, Arusha, Tanzania
- 2010 **Masatoshi Katabuchi**, Hiroko Kurokawa, Sylvester Tan, and Tohru Nakashizuka, “Soil variation and functional trait assembly in a diverse dipterocarp forest”. 95th ESA Annual meeting, Pittsburgh, USA
- 2009 **Masatoshi Katabuchi**, Jun Yokoyama, and Tohru Nakashizuka, “Inbreeding and mating structure in a dioecious fig pollinating wasp”. ATBC, Marburg, Germany

## Outreach

- 2015 Masatoshi Katabuchi “Leaves and tropical rainforests”. Invited by the Japanese Association of Interdisciplinary Study Group in University Florida

## — Languages

Japanese (native), English (fluent), Malay (basic), Chinese-Mandarin (basic)

## — Graduate Students

Zhou Cong, 2021 - Present, M.Sc. Student, Recipient of UCAS Scholarship.